KSP Datathon: Project Netra Cloud Blueprint

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Part 1: Required Software Apps to Install

Before writing a single line of code, you must download and install these desktop applications on your laptop:

- **Visual Studio Code (VS Code):** The code editor where you will type all your Python scripts.
- **DB Browser for SQLite:** A visual tool to open and inspect the fake cybercrime database you generate.
- **Postman:** An app to test your Catalyst cloud endpoints and GLM 4.7 API connections before wiring them to your frontend.
- **Node.js:** Required only because the Zoho Catalyst Command Line Interface (CLI) needs it to run on your laptop.

Part 2: Deep Step-by-Step Execution Plan

Step 1: Generate Synthetic Police FIR Data

Actionable Steps:

1. Open VS Code and create a new folder called `Project_Netra`.
2. Open the terminal in VS Code and install the required libraries by typing: `pip install faker pandas`.
3. Copy the AI prompt below, paste it into an AI, and put the generated Python code into a file named `generate_data.py`.
4. Run the script. It will output multiple CSV files.
5. Open DB Browser for SQLite, import your new CSV files, and visually verify that the 1.5 Lakh rows of FIR, Accused, Victim, and CrimeHead data look realistic according to the KSP schema.

AI Prompt: Data Generation
"Act as a Python Data Engineer. Write a script using the 'Faker' library and Pandas to generate 150,000 synthetic records strictly based on the Karnataka Police FIR System schema. Create interconnected relational datasets for 'CaseMaster', 'Victim', 'Accused', 'Complainant Details', and 'Arrest Surrender'. Ensure the 'CrimeNo' in 'CaseMaster' perfectly follows this format: 1 digit Case Category Code + 4 digit District ID + 4 digit Police Station ID + 4 digit Year + 5 digit Running Serial Number. Use lookup tables for 'CrimeHead', 'CrimeSubHead', and 'Gravity Offence' and maintain proper Foreign Key relationships. Save outputs as interconnected CSV files."

Step 2: Build the Agentic AI Core (GLM 4.7 Flash)

Actionable Steps:

1. In your VS Code terminal, install the AI frameworks: `pip install langchain sqlalchemy`.
2. Configure your Python script to consume the native **GLM 4.7 Flash** model provided by the Zoho Catalyst platform for your LangChain agent.
3. Use the AI prompt below to generate the code for a custom LangChain agent that interfaces with GLM 4.7 Flash. Paste it into `ai_agent.py`.

AI Prompt: Agentic Core
"Act as an AI Architect. Write a Python script using LangChain's 'create_sql_agent'. Connect it to a SQLite database containing the exact Karnataka Police FIR System schema data. Configure the LLM connection to consume the GLM 4.7 Flash model API natively provided by Zoho Catalyst. Ensure the agent converts a natural language query into SQL, executes it across the relational tables (such as CaseMaster and Accused), and returns a logical summary."

Step 3: Integrate Kannada Voice & Text via Catalyst

Actionable Steps:

1. Bypass the Bhashini API to avoid administrative delays. Use the open-source OpenAI Whisper model for audio processing.
2. Create a Python script that takes audio input, uses Whisper to transcribe Kannada/Kanglish to text, and natively routes it to Catalyst's GLM 4.7 Flash to translate it into strict English.
3. Integrate this translation pipeline directly into your `ai_agent.py` file so it acts as a pre-processor before the SQL search.

AI Prompt: NLP Integration
"Act as an NLP Engineer. Write a Python function using the open-source Whisper library to transcribe Kannada/Kanglish audio files. Then, write a system prompt for the GLM 4.7 Flash model to accurately translate that code-mixed transcript into clean English for SQL database processing."

Step 4: Deploy on Zoho Catalyst Cloud Scale

Actionable Steps:

1. Open your terminal and install the Zoho CLI by typing: `npm install -g zcatalyst-cli`.
2. Type `catalyst login` in the terminal to connect your VS Code to your Zoho account.
3. Upload your SQLite database or CSV files into the Catalyst Relational Data Store via their web dashboard.
4. Deploy your Python code as a Catalyst Serverless Function since the LLM models (GLM 4.7 and Qwen 3.6) are already hosted by the platform natively.

AI Prompt: Deployment
"Act as a Cloud DevOps Engineer. Write the terminal steps and folder structure required to deploy a Python-based LangChain agent as a Zoho Catalyst Serverless Function using the `zcatalyst-cli`."

Kanglishnalli Vivaravada Mahiti (Deep Explanation in Kanglish)

Geleya, neevu kelidante ee project annu madalu bekaada apps mattu exact aagi enu madabeku annuva step-by-step vivarane illide:

Bhaga 1: Install madabekada Apps

Neevu code bareyokke munche nimma laptop nalli ee apps irabeku:

- **VS Code:** Illi neevu Python code type maduttira.
- **DB Browser:** Neevu srushtisida nakali data sariyagideya anta open madi nodokke.
- **Postman:** Cloud endpoints mattu GLM 4.7 sariyagi kelsa maduttideya anta test madokke.
- **Node.js:** Zoho CLI command galu nimma laptop nalli kelsa madalu idu beku.

Bhaga 2: Step-by-Step Action Plan

Hanta 1: Data Srushti (Data Generation)
VS Code open madi faker mattu pandas install madi. AI kotta code annu paste madi run madi. Idu CSV files koduttade. Aa file annu DB Browser nalli open madi, 1.5 laksha FIR mattu crime data KSP schema prakara sariyagideya anta check madi.

Hanta 2: AI Medulu Nirmana (GLM 4.7 Flash Core)
Zoho Catalyst native aagi kodutiruva GLM 4.7 Flash model annu balasi. VS Code nalli LangChain install madi. AI prompt balasi bandiruva code annu paste madi run madi. Idu nimma SQL database jote direct aagi mathaduttade.

Hanta 3: Kannada/Kanglish Alavadike
Bhashini bittubidi. Open-source Whisper model balasi Kanglish audio text ge convert madi. Nantara Catalyst nalli iruva GLM 4.7 Flash balasi English ge translate madi. AI kotta Python code annu nimma main code jote seri.

Hanta 4: Zoho Catalyst nalli Hakuvudu (Deployment)
Terminal nalli `npm install` command balasi Zoho CLI install madi. `catalyst login` madi nimma account connect madi. Zoho website ge hogi nimma CSV data upload madi. Heavy Docker containers beda, direct aagi Catalyst Serverless Functions moolaka nimma project annu deploy madi.

All the best geleya, neevu idannu khandita gelluttira!